

Northwind Interactive Sales Dashboard Using Power BI

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INTRODUCTION:

The Northwind dataset contains sales and business information that can be used to understand order activity, customer distribution, product categories, and overall sales performance. Analysing this data helps identify important patterns such as changes in the number of orders over time, sales generated by different product categories, and the distribution of customers across countries.

Microsoft Power BI is used in this project to transform the Northwind data into an interactive sales dashboard. The dashboard presents key information through charts, KPI cards, and filters, covering areas such as total orders, total sales, quantity sold, freight, customers, countries, and product categories. These visualizations make the data easier to understand and compare.

The main purpose of this project is to develop an Interactive Northwind Sales Dashboard that provides a clear overview of sales performance. The dashboard also includes interactive slicers and a drill through feature, allowing users to filter the information and move from summarized data to more detailed information. This demonstrates how Power BI can be used to explore business data and support effective data-driven analysis.

OBJECTIVES OF THE PROJECT:

- To analyse the Northwind sales data using Microsoft Power BI.
- To evaluate key indicators such as Total Orders, Total Sales, Quantity Sold, Total Freight, and Number of Customers.
- To analyse the monthly trend of total orders and identify changes in order activity over time.
- To compare customer distribution across different countries.
- To analyse sales performance across different product categories.
- To provide interactive filtering using Year, Region, and Category Name slicers.
- To implement a drill through feature for accessing detailed information from the main dashboard.
- To create an interactive dashboard that presents meaningful sales and customer insights in an easy-to-understand format.

DATASET DESCRIPTION:

The Northwind dataset contains information related to orders, customers, products, categories, employees, and shipping details. It provides useful business data for analysing sales performance, customer distribution, order activity, and

product categories. The dataset can be used to identify sales trends and understand different aspects of the company's operations.

The dataset was imported into Microsoft Power BI for data analysis and visualization. The information was used to create KPI cards and visualizations showing total orders, total sales, quantity sold, freight, customer count, country-wise customers, and category-wise sales.

Key Fields in the Dataset

- Order ID – Unique identification number for each order.
- Order Date – Date on which the order was placed.
- Customer ID – Unique identification of each customer.
- Customer Name – Name of the customer.
- Country – Country associated with the customer or order.
- Product Name – Name of the product included in an order.
- Category Name – Category to which the product belongs.
- Quantity – Number of units ordered.
- Unit Price – Price of each product unit.
- Freight – Shipping and transportation cost associated with an order.
- Region – Geographical region associated with the data.

DATA PREPARATION:

Before developing the dashboard, the Northwind dataset was prepared in Power BI to ensure that the data was suitable for analysis and visualization.

- Imported the Northwind dataset into Power BI Desktop.
- Checked and corrected the data types of date, numerical, and text fields.
- Reviewed fields such as Order Date, Quantity, Unit Price, and Freight for accurate calculations.
- Checked customer, product, category, and country information for consistent analysis.
- Organized the relevant fields for creating KPI cards, charts, and slicers.
- Prepared the data for monthly, country-wise, and category-wise analysis.
- Set up the required fields to support the drill through functionality for detailed analysis.

This preparation ensured that the Northwind data was organized and ready for creating an interactive Power BI dashboard.

DAX MEASURES USED:

- Total Orders – Calculates the total number of orders recorded in the dataset.

- Total Sales – Calculates the overall sales value generated from the orders.
- Total Quantity – Calculates the total number of products ordered.
- Total Freight – Calculates the total freight or shipping cost associated with the orders.
- Customer Count – Calculates the number of customers included in the dataset.

Dashboard Components

The Northwind Sales Dashboard consists of KPI cards, charts, slicers, and a drill through feature designed to provide an interactive view of sales performance. Each component focuses on a specific aspect of the dataset, including orders, sales, quantity, freight, customers, countries, and product categories. Together, these components make it easier to identify sales trends, compare business performance, and explore detailed information from the dashboard.

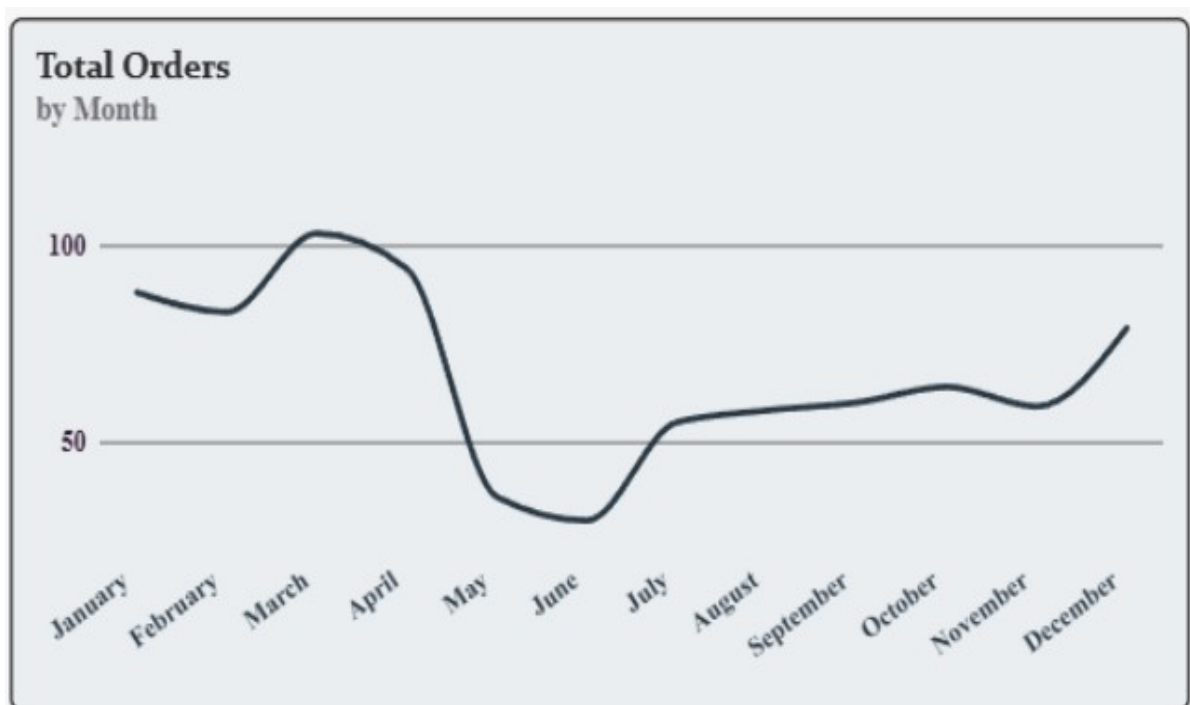
✓ Total Orders by Month

A line chart is used to display the number of orders recorded across different months. The months are represented on the X-axis, while the total number of orders is represented

on the Y-axis.

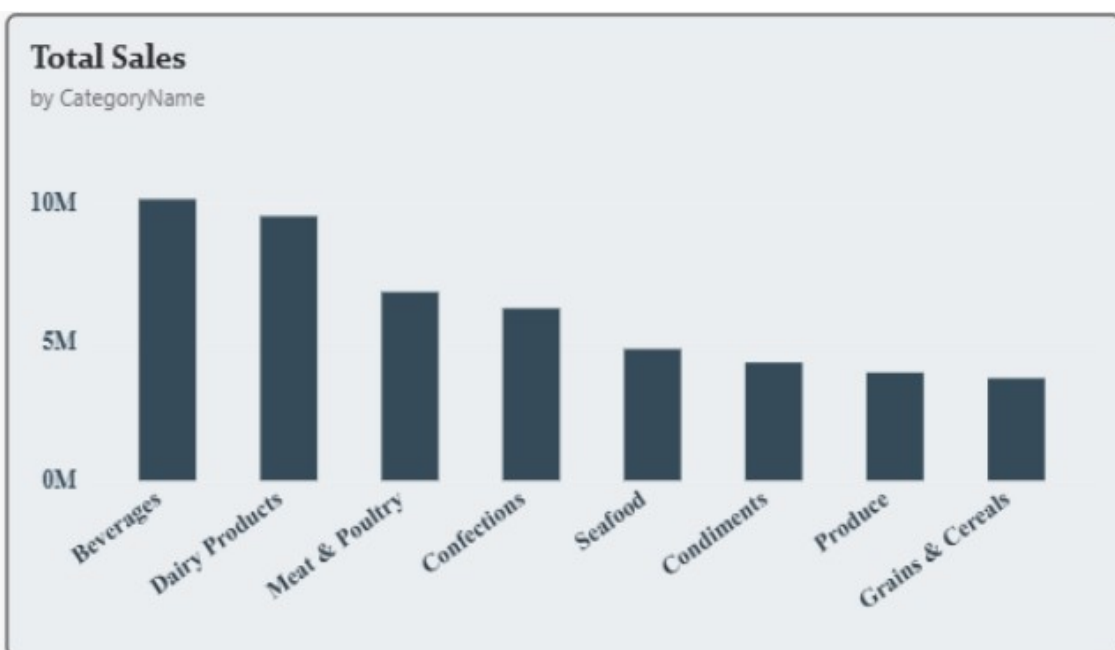
- The chart shows the monthly trend in total orders throughout the year.
- Orders increase from January and reach their highest level around March.
- A noticeable decline occurs from April to June, with June showing the lowest order level.
- Orders begin to recover from July onwards.
- The trend shows a gradual increase through the later months, followed by a rise in December.
- The line chart makes it easy to identify increases, decreases, and seasonal variations in order activity.

✓ **Total Sales by Category**



A column chart is used to compare the total sales generated by different product categories. The Category Name is represented on the X-axis, while Total Sales is represented on the Y-axis.

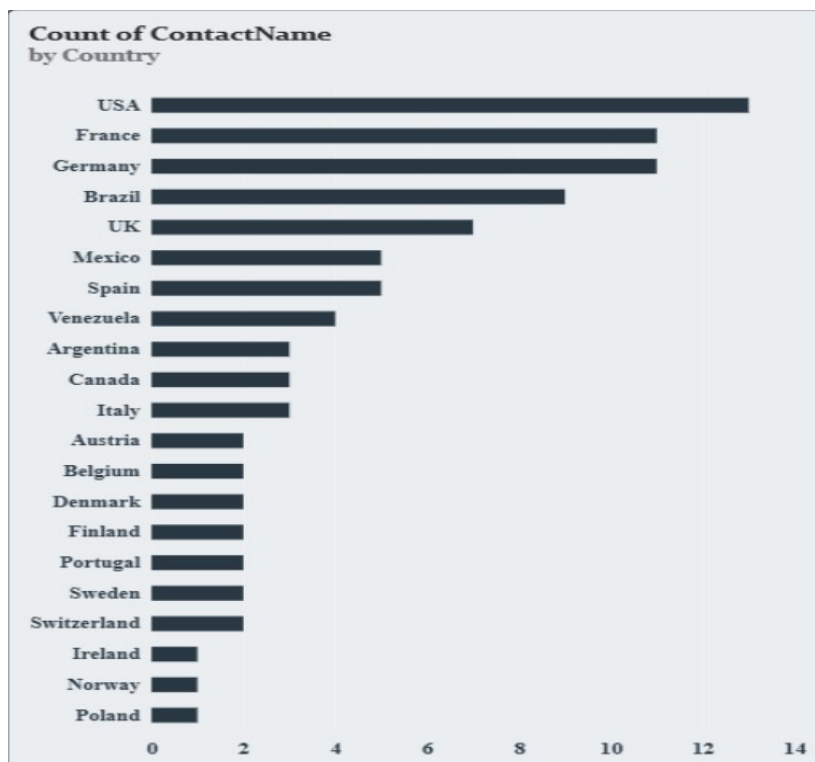
- Beverages record the highest total sales at approximately 10M.
- Dairy Products follow with sales of approximately 9.5M.
- Meat & Poultry generate around 6.7M in sales.
- Confections contribute approximately 6.1M.
- Seafood, Condiments, Produce, and Grains & Cereals show comparatively lower sales values.
- The chart clearly highlights the differences in sales performance across product categories.



✓ Count of Customers by Country

A horizontal bar chart is used to display the number of customers across different countries. The country names are shown on the Y-axis, while the customer count is represented the X-axis.

- The USA has the highest customer count, with approximately 13 customers.
- France and Germany follow with around 11 customers each.
- Brazil has approximately 9 customers, followed by the UK with around 7 customers.
- Mexico and Spain have around 5 customers each.
- Countries such as Ireland, Norway, and Poland have comparatively fewer customers.



- The chart provides a clear comparison of customer distribution across different countries.

SLICERS USED

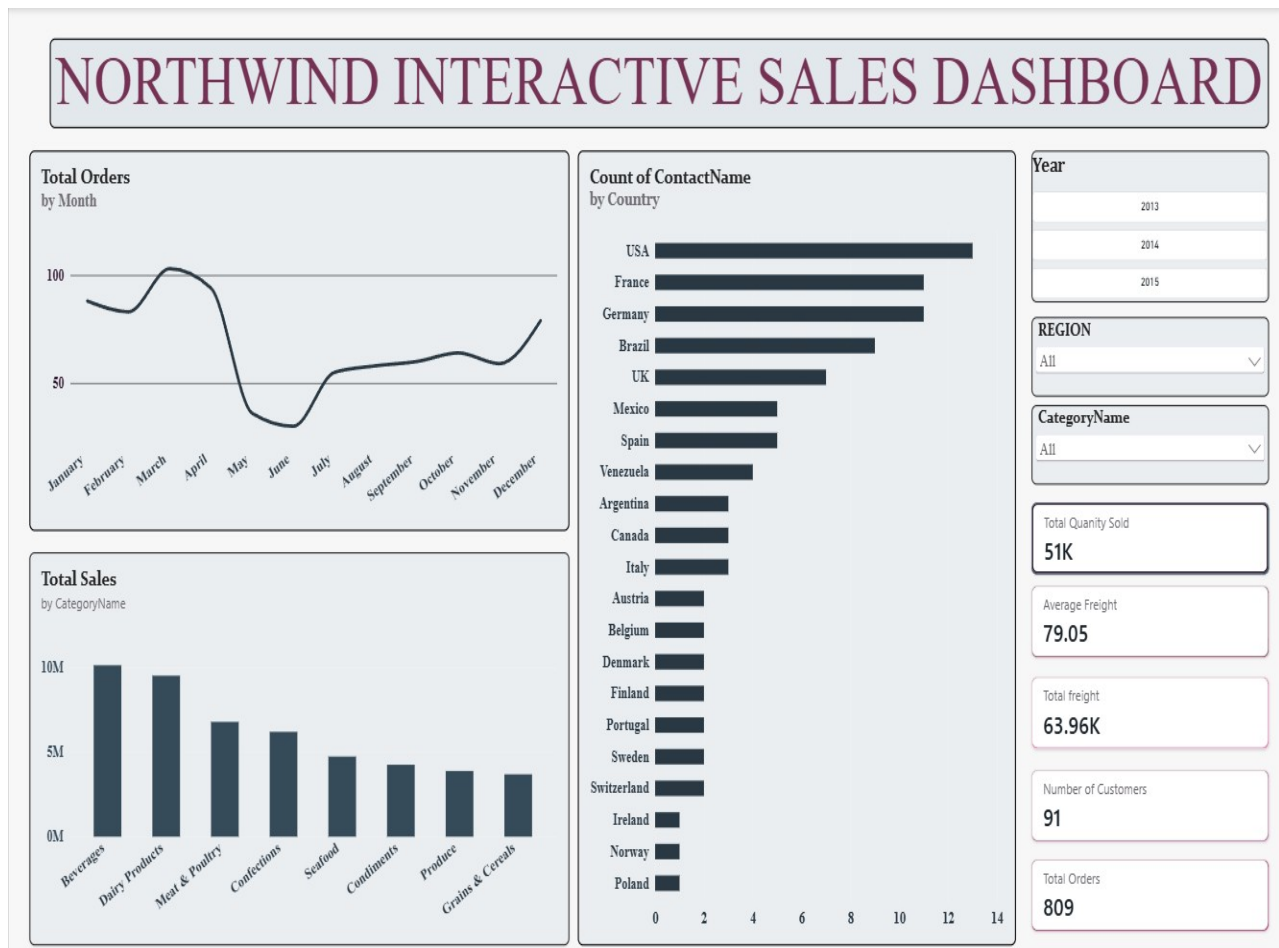
The dashboard uses three slicers to make the analysis interactive. These slicers allow users to filter the dashboard and view information based on selected criteria.

- **Year Slicer** – Allows users to select a specific year, such as 2013, 2014, or 2015. This helps analyse sales and order performance for a particular year.
- **Region Slicer** – Allows users to select a specific region or view all regions. This helps compare sales and customer performance across different geographical regions.
- **Category Name Slicer** – Allows users to select a particular product category or view all categories. This helps analyse

The image displays three vertical slicers. The top slicer, titled 'Year', is a list box with three options: 2013, 2014, and 2015. The middle slicer, titled 'REGION', is a dropdown menu currently showing 'All' with a downward arrow. The bottom slicer, titled 'CategoryName', is also a dropdown menu currently showing 'All' with a downward arrow.

sales and order information for selected product categories.

Final Dashboard



The final Northwind Interactive Sales Dashboard combines KPI cards, charts, and slicers into a single interactive report. It provides an overview of total orders, sales performance, customer distribution, quantity sold, freight, and product categories.

The dashboard allows users to analyse monthly order trends, compare sales across product categories, and examine customer distribution by country. The Year, Region, and Category Name slicers provide interactive filtering, while the

drill through feature allows users to move from summary-level information to more detailed data.

Overall, the final dashboard presents the Northwind sales data in a clear and interactive format, making it easier to explore business performance and identify important sales and customer patterns.

CONCLUSION:

The Northwind Interactive Sales Dashboard demonstrates how Power BI can transform sales data into clear and meaningful insights. It helps analyse orders, sales, customers, and product categories through interactive visuals, slicers, and drill through features.